Tayyab Tahir

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Research Profile

Reinforcement Learning | Data-Driven & Predictive Control | Trustworthy AI | Microgrids, V2G & VPPs | Field Robotics

Electrical engineer working on reinforcement learning and predictive control for energy systems under uncertainty: microgrids with hydrogen storage, vehicle-to-grid, virtual power plants. I build the simulators, benchmark deep RL against model predictive control (MPC) and rule-based baselines, and report where learned controllers fail. At TalTech I work on trustworthy AI for field robots: prediction intervals with coverage checks for rover wheel-slip estimation; deep-RL traction and locomotion control is ongoing. Nine peer-reviewed papers, two in *Applied Energy*.

Research interests: multi-agent RL for distributed energy resources; learning-based MPC; trustworthy and safe RL; sim-to-real robot locomotion.

Education

Erasmus Mundus Joint Master in Marine and Maritime Intelligent Robotics (MIR), 2023 – 2025, full scholarship

Norwegian University of Science and Technology (NTNU) **Aug. 2024 – Aug. 2025**
MSc Marine Robotics (Erasmus Mundus MIR), Electrical, Computer & Control Eng. | *Grade: A* *Trondheim, Norway*

- Thesis: *Modeling and Control of Smart Integrated Port Energy Systems: A Data-Driven Approach*. Supervisor: Prof. Mehdi Zadeh.
- Courses: Control Systems, Marine Control Systems, Hybrid Power Systems, Machine Learning.

Université de Toulon **Sep. 2023 – Jul. 2024**
MSc Engineering of Complex Systems (Erasmus Mundus MIR) | *Grade: 15.2/20* *La Garde, France*

University of Management and Technology (UMT) **Oct. 2016 – Nov. 2020**
BS Electrical Engineering | *GPA: 3.96/4.00, Gold Medalist* *Lahore, Pakistan*

Research & Professional Experience

TalTech, Centre for Biorobotics **Dec. 2025 – Present**
Early Stage Researcher, Horizon Europe project *Tallinn, Estonia*

- Lead developer of GCAST, an uncertainty-aware transformer that estimates rover wheel slip from multi-rate IMU and wheel-odometry streams without resampling them to a common rate (Demo).
- Trustworthy AI: prediction intervals from the transformer (heteroscedastic head, MC-Dropout), evaluated for coverage and against gradient-boosting quantile regression and Gaussian processes; tested under scarce labels and sensor dropout on three public rover datasets.
- Terrain-classification branch compared against published CNN and Mamba baselines on BorealTC. Journal paper in preparation (*Eng. Appl. of Artificial Intelligence*).
- ROS 2 skid-steer kinematics and odometry for the project rovers; developing a deep-RL traction and locomotion controller (in progress).
- Tool-using LLM agents (MCP) for the robot platform; SLURM/HPC training pipelines; Horizon deliverables.

NTNU, Department of Marine Technology **Jan. 2025 – Jun. 2025**
MSc Thesis Researcher, Smart Port Energy Systems *Trondheim, Norway*

- Gymnasium environment of a port microgrid (PV, 5.5 MWh battery, electrolyser, H_2 storage, fuel cell, grid) on hourly 2023 data.
- Benchmarked DQN, PPO, MPC (CVXPY) and an LSTM trained to imitate MPC; the imitation policy stayed within 2% of MPC cost at about 250× faster inference.
- Oslo simulation: MPC–LSTM cut the electricity bill by 38.9% and CO_2 by 45.8% against the non-optimised base case.
- Same policy at three other ports without retraining exposed hydrogen-storage saturation and emission increases where carbon intensity was missing from the state (Demo | Code).

NTNU, Department of Marine Technology **Jan. 2025 – Jun. 2025**
Teaching Assistant, Marine Electric & Hybrid Power and Propulsion Systems *Trondheim, Norway*

- Simulink labs on DC distribution, hybrid power and converter control.

EXAIL (formerly ECA Group) **Jun. 2024 – Aug. 2024**
AI Engineering Intern *La Garde, France*

- Built a retrieval-augmented LLM assistant over the company's internal codebase and documentation.

UMT, Smart Power Systems Lab **Feb. 2021 – Aug. 2023**
Lab Engineer *Sialkot, Pakistan*

- PV–battery energy management and MPPT, validated in real-time hardware-in-the-loop tests.
- Microgrid testbed: ESP32/STM32 control, MQTT and Modbus telemetry into SCADA with anomaly alerts.

Thunder Solar Solution **Jan. 2020 – Feb. 2021**
Solar PV Design & Commissioning Engineer *Sialkot, Pakistan*

- Designed and commissioned 15+ grid-tied and hybrid PV systems (3–150 kWp).

Selected Research Projects

Virtual Power Plant Control with Deep RL | *PyTorch, SAC, OMIE market data, SLURM* | Demo **2025 – 2026**

- PV, wind, battery and EV fleet doing day-ahead arbitrage and frequency reserve. Year-long simulation (in-sample prices): SAC +71% net value vs rule-based control, no EV charge violations; held-out week, SAC vs rule-based: FCR delivery 0.94 vs 0.73, net value EUR 219 lower.

Lumen Home: Home Energy Digital Twin | *TypeScript, dynamic programming, PVGIS, Nord Pool* | Demo **2026**

- PV–battery–EV (V2H)–heat-pump simulator; 36 h dynamic programming on P50 forecasts. Tallinn simulation vs the same house on default rules: 35% lower annual bill; blackout autonomy 2.3 to 19.3 h with load shedding.

City Blackout Resilience Planner | *Python, MILP (HiGHS), pandapower, FastAPI* | City Resilience Hack, Tallinn **2026**

- 72 h rolling-horizon MILP grouping EV fleets, batteries and generators into energy islands on a city-scale grid twin.

GlassML: Glass-Box ML Testbench | *Next.js, FastAPI, PyTorch, scikit-learn, Stable-Baselines3* | Demo **2026**

- Self-hosted platform to profile data, build models visually and run supervised, deep and RL experiments with live diagnostics.

Marine Robotics: BlueROV Servoing & Dynamic Positioning | *ROS, OpenCV, Simulink* | Project page **2024 – 2025**

- Team projects; my part: OpenCV buoy detection, visual-servo control law (tank-tested); DP PID/LQG control, wave/wind simulation.

Publications

9 peer-reviewed papers (5 journal, 4 conference); 168 citations, h-index 6 (Google Scholar, Oct. 2026). My part in each of [1]–[3], [6], [7]: deep-RL/control layer, EV battery model, battery-limit reward penalties, cluster training, methodology; [4] Simulink plant, PSO tuning, ANN.

[1] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "Novel artificial intelligence driven model-based control framework for solar–electric vehicle home energy optimization," *Engineering Applications of Artificial Intelligence*, vol. 177, 114868, 2026.

[2] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "Novel control-based design optimization of smart energy distribution and management in vehicle-to-grid integrated microgrid," *Applied Energy*, vol. 402, 127047, 2026.

[3] M. Irfan, **T. Tahir**, S. Huang, S. Deilami, B. P. Veettil, "Optimizing residential energy costs: A novel machine learning approach for solar PV and EV integration through heuristic price signal dispatch," *IEEE Transactions on Industry Applications*, vol. 61, no. 3, pp. 4684–4694, 2025.

[4] M. Irfan, S. Deilami, S. Huang, **T. Tahir**, B. P. Veettil, "Optimizing load frequency control in microgrid with vehicle-to-grid integration in Australia: Based on an enhanced control approach," *Applied Energy*, vol. 366, 123317, 2024.

[5] **T. Tahir**, A. Khalid, J. Arshad, A. Haider, I. Rasheed, A. U. Rehman, S. Hussen, "Implementation of an IoT-based solar-powered smart lawn mower," *Wireless Communications and Mobile Computing*, vol. 2022, 1971902, 2022.

[6] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "A novel predictive strategy for daily energy cost reduction in solar 'PV + EV' system with V2G and V2H modes," *IEEE ETFG*, 2025.

[7] M. Irfan, **T. Tahir**, S. Deilami, S. Huang, B. P. Veettil, "A novel data-driven optimization for cost-effective home energy management with PV–EV integration," *IEEE AUPEC*, 2024.

[8] Z. M. Tahir, **T. Tahir**, et al., "Design and implementation of prepaid energy meter with home automation and peak load management using Android application," *Engineering Proceedings* (IEEC 2023), 2023.

[9] A. Khalid, **T. Tahir**, et al., "Autonomous vehicle based on lane detection through computer vision using Raspberry Pi," *ICIT*, 2022.

In preparation: uncertainty-aware transformer for rover wheel-slip estimation. Full list: tayyab-tahir1.github.io/publications.

Technical Skills

Reinforcement Learning: deep Q-network (DQN), proximal policy optimisation (PPO), soft actor-critic (SAC), imitation learning, SB3
Control & Optimisation: MPC (CVXPY), MILP (PuLP, HiGHS, Pyomo), dynamic programming, PID/LQG, EKF and observers
Trustworthy AI: prediction intervals and coverage, MC-Dropout, quantile gradient boosting, Gaussian processes, conformal
Machine Learning: PyTorch, transformers, LSTM, Mamba, scikit-learn, XGBoost/LightGBM, Optuna
Energy Systems: microgrids, V2G/V2H, VPPs, energy management systems (EMS), PV–battery–H_2; PyPSA, pandapower, HIL
Robotics: wheeled mobile robots, wheel-slip estimation, skid-steer locomotion, ROS / ROS 2, visual servoing, OpenCV, ESP32/STM32
Research Computing: Python, C/C++, MATLAB, TypeScript, SLURM/HPC, Docker, Git, W&B, LaTeX, LLM agents, MCP

Certifications

NASA Open Science 101, NASA Science Mission Directorate, Aug. 2026 | Credly badge

Building with the Claude API, Anthropic Academy, Apr. 2026

Introduction to Model Context Protocol, Anthropic Academy, Aug. 2026

Honours & Awards

- Gold Medalist, BS Electrical Engineering, UMT; Rector Scholarship and five Rector Awards, 2016 – 2020.
- NGIRI National Award 2020 (Ministry of IT & Telecom, Pakistan): final-year project funded nationally.

Languages

English: PTE Academic 70 (Aug. 2026, valid to Aug. 2028) | Urdu: native

References

References available on request.